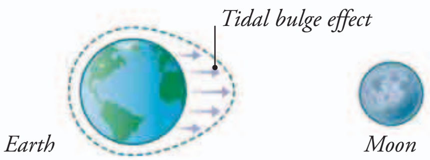
Tides

On most seashores the sea level rises and falls every day, flooding part of the shore and then exposing it again. These high and low tides are caused by the gravity of the Moon, modified by other forces. The tidal rise and fall also creates strong local currents that change direction every few hours.

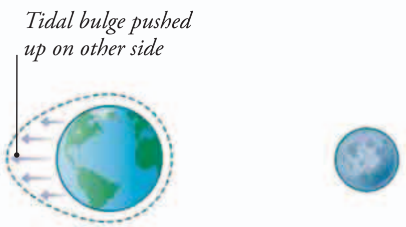
▲ LOW TIDE
This seashore in Vietnam experiences just one cycle of high and low tide every 24 hours, unlike most shores around the world.

PULLED BY THE MOON

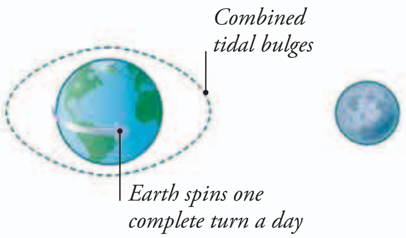
Shown below is the way the Moon's gravity pulls on the oceans, and drags the water into two vast tidal bulges. As Earth spins on its axis, most of its shores pass in and out of these bulges, causing high and low tides.



▲ GRAVITY EFFECT
Ocean water is dragged toward the Moon by the force of gravity, so sea level rises on the side facing the Moon.



▲ MIRROR IMAGE
Earth is also orbiting the Moon very slightly, and this creates a second tidal bulge on the side of Earth facing away from the Moon.



▲ SPINNING EARTH
The bulges stay in line with the Moon, so as Earth spins, any one seashore will pass through two tidal bulges each day.

TIDAL SHORES

When the tide level falls, it exposes parts of the shore that have been covered with seawater for several hours. On rocky shores with steep cliffs, the difference may not be very obvious, but on shallow-sloping shores such as this sandy beach, a fall of several feet can expose a vast area of tidal flat.

▲ FLOOD TIDE
When the tide level starts rising again, seawater floods back to cover the sand. This can happen very quickly, transforming the beach into a glittering expanse of shallow sea.